Mastering Diffusion

Week 6 – Various Diffusion Space

Latent Diffusion Models and Discrete Space Diffusion Models

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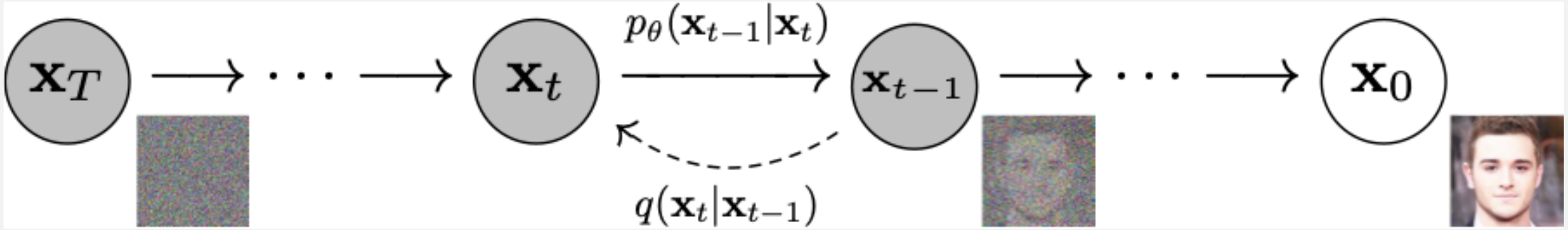
AIDAS Lab

Latent Space Diffusion

Recap: Diffusion Models

- DDPM trains with forward and backward process

$$q(x_{1:T}|x_0) := \prod_{t=1}^{T} q(x_t|x_{t-1})$$



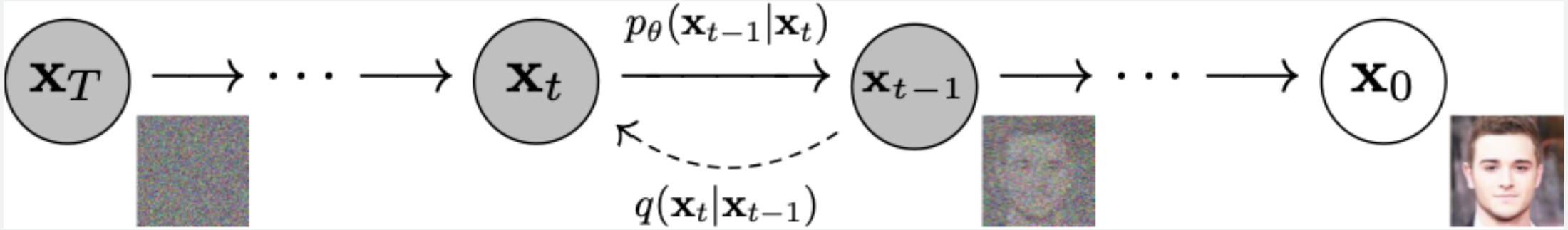
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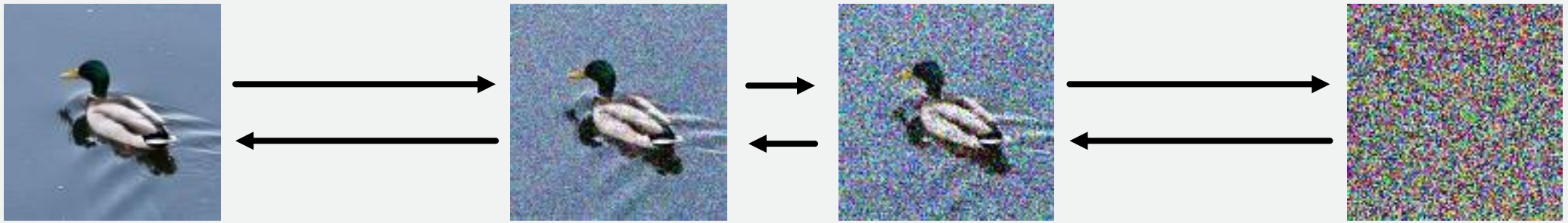
$$p_\theta(\boldsymbol{x}_{0:T}) := p(\boldsymbol{x}_T) \prod_{t=1}^{T} p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$$



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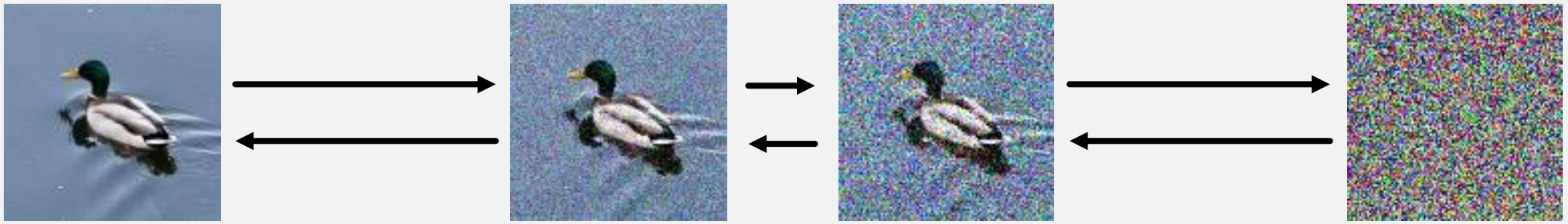
DDPM Implementation

Forward Process: $q(x_t|x_{t-1}) \coloneqq \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t \mathrm{I})$



DDPM Implementation

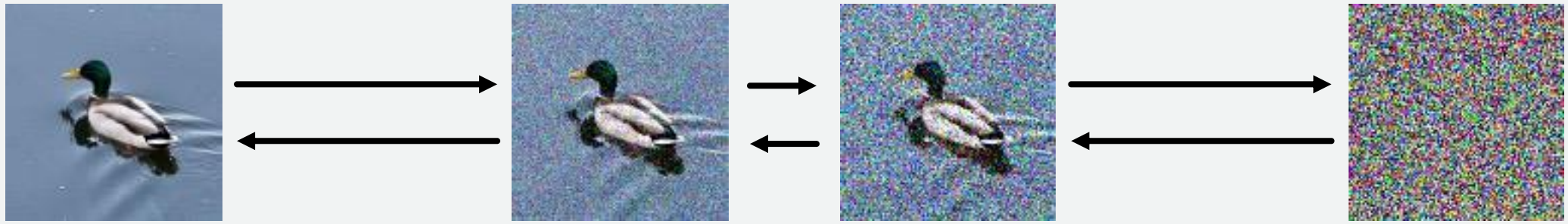
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Backward Process: $p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) \coloneqq \mathcal{N}(\boldsymbol{x}_{t-1}, \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t))$

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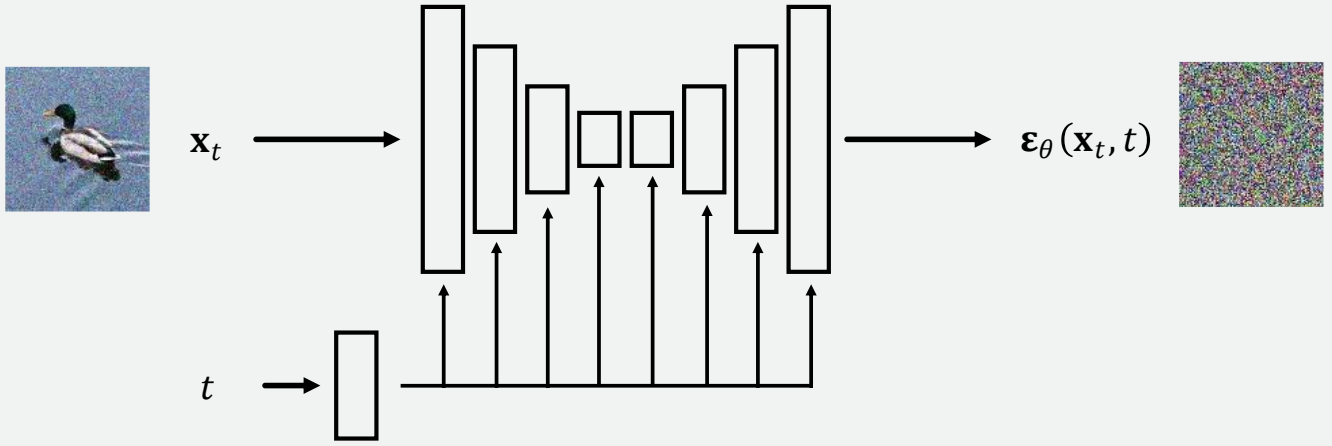
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Recall (Week 4): The only thing the model needs to predict is the epsilon

$$\boldsymbol{\mu_\theta}(\boldsymbol{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha_t}}}\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\right)$$

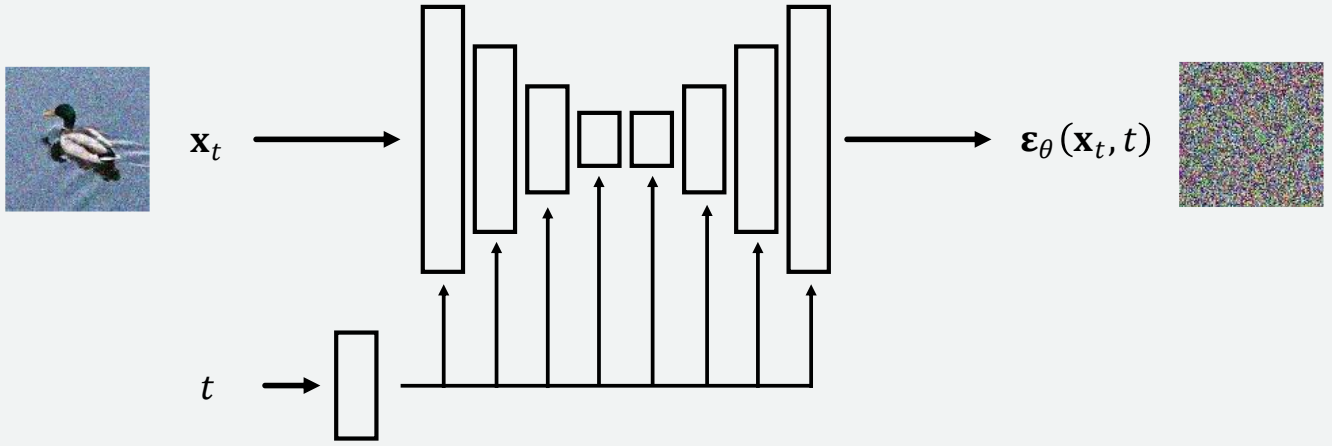
DDPM Implementation

- To predict the ϵ, Diffusion models often use the U-Net architecture



DDPM Implementation

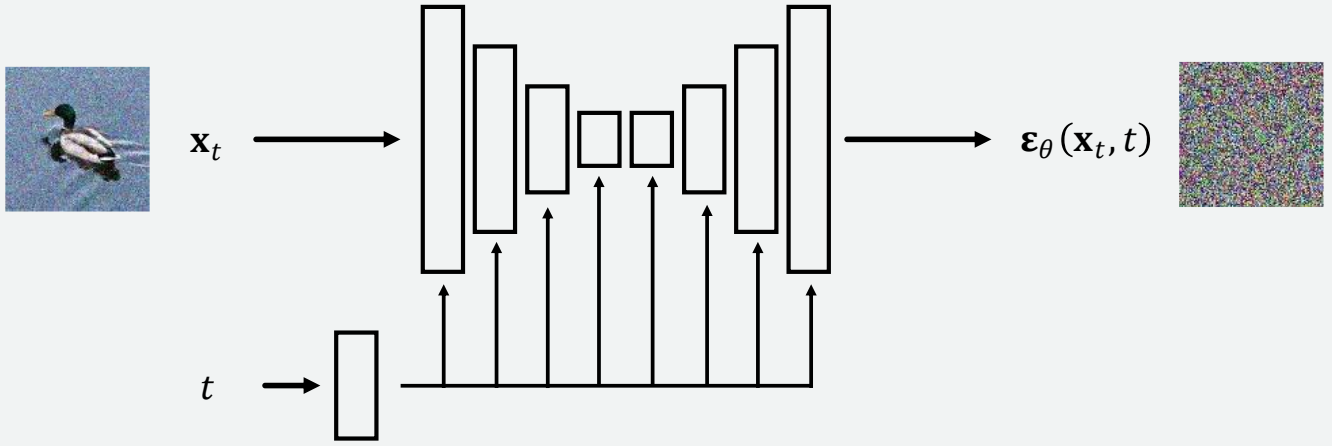
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DDPM Implementation

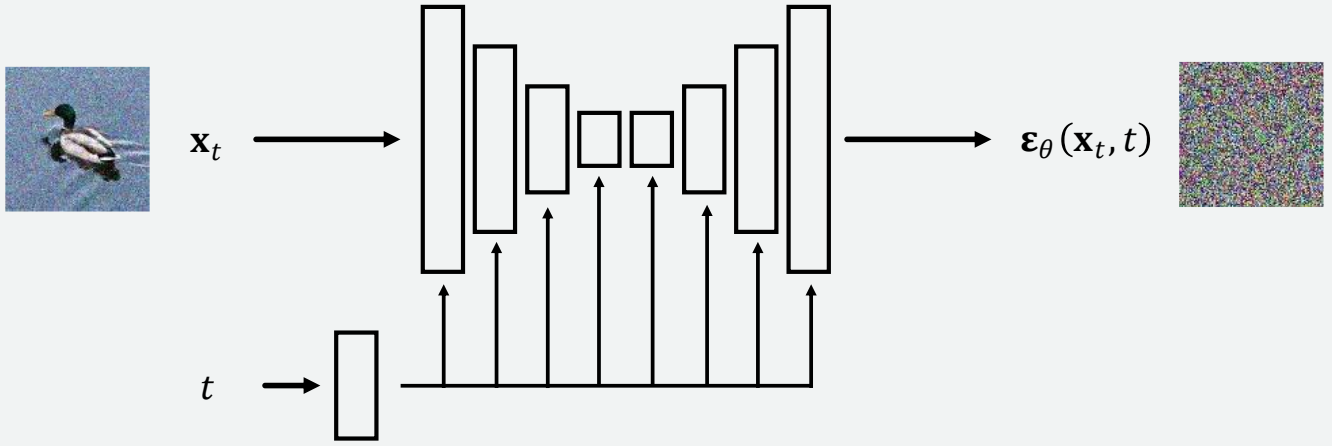
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DDPM Implementation

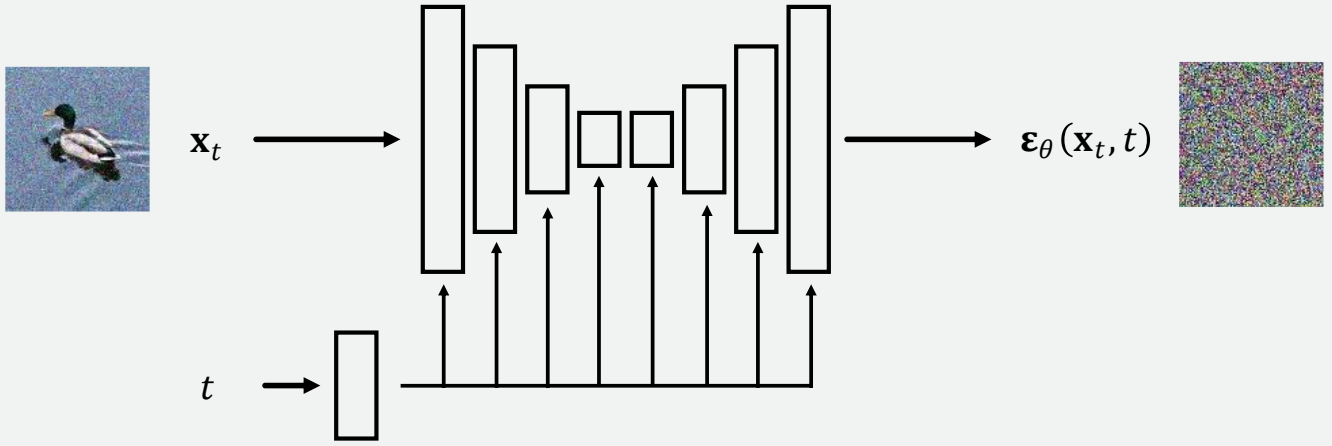
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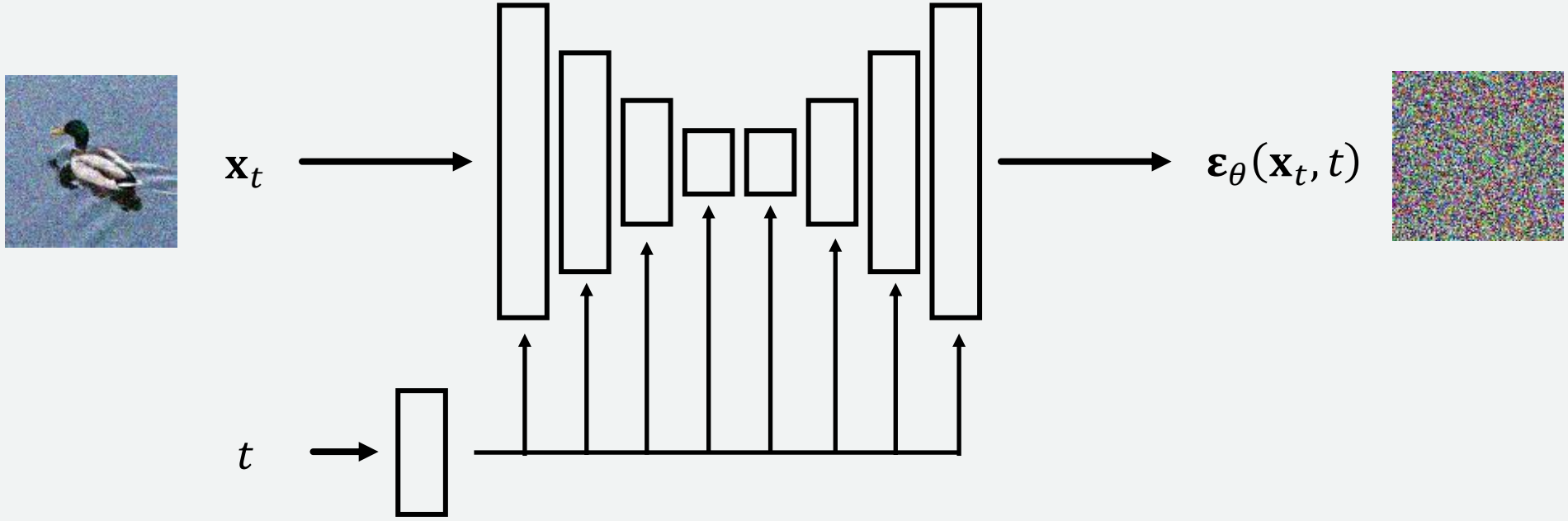
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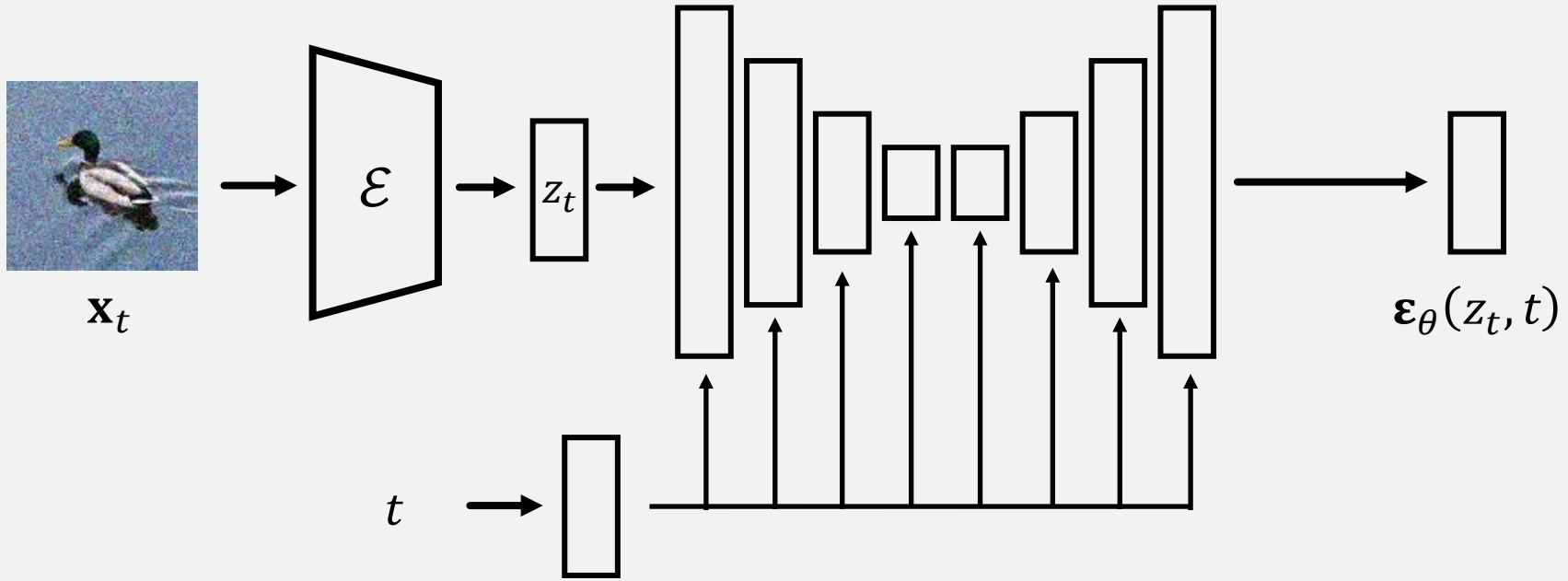
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-> Reduce the dimension by predicting p(x) in latent space

Latent Diffusion Model

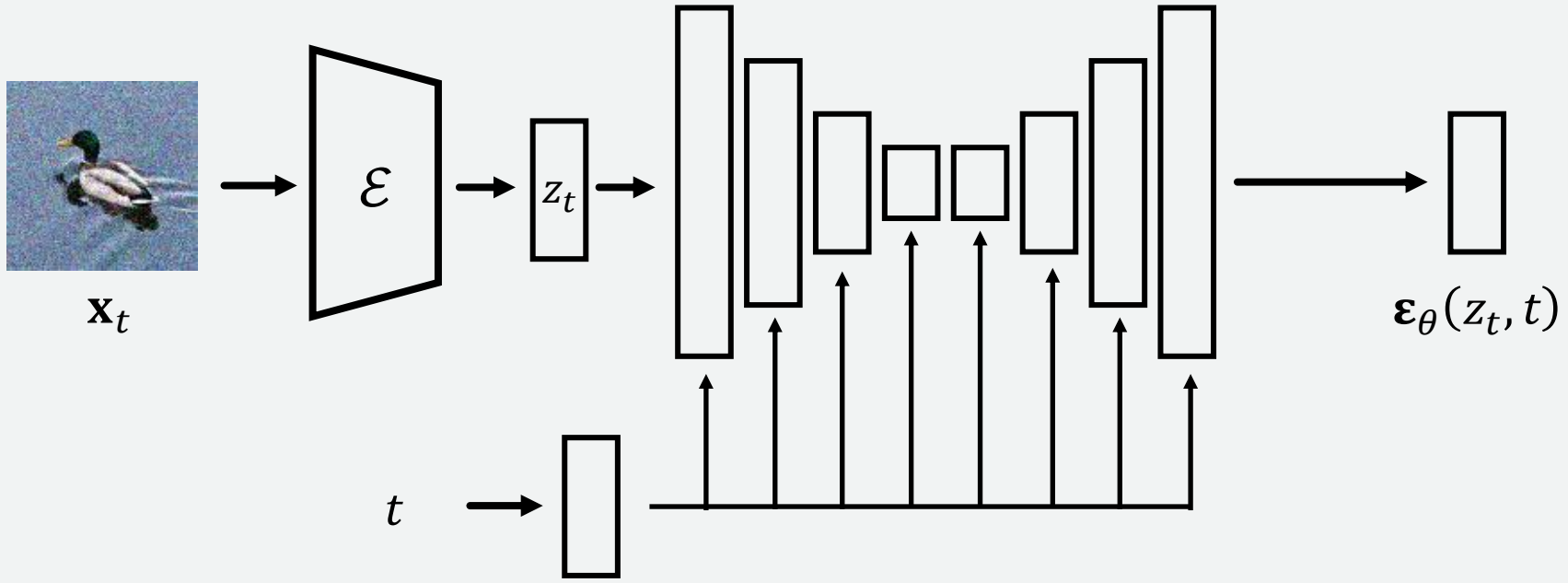


Latent Diffusion Model



R. Rombach et al., “High-Resolution Image Synthesis with Latent Diffusion Models”, CVPR 2022.

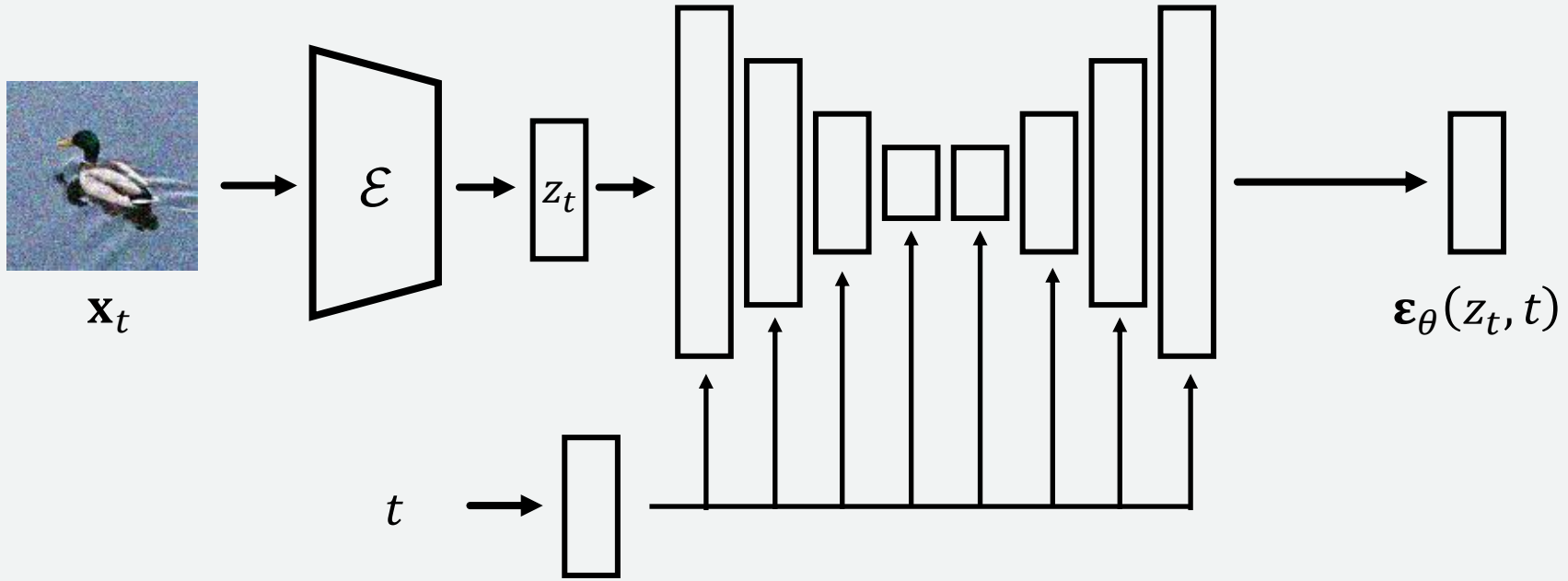
Latent Diffusion Model



$$\mathcal{L}_{LDM} := \mathbb{E}_{\mathcal{E}(x),\epsilon\sim\mathcal{N}(0,1),t}[\|\varepsilon - \varepsilon_\theta(z_t, t)\|_2^2]$$

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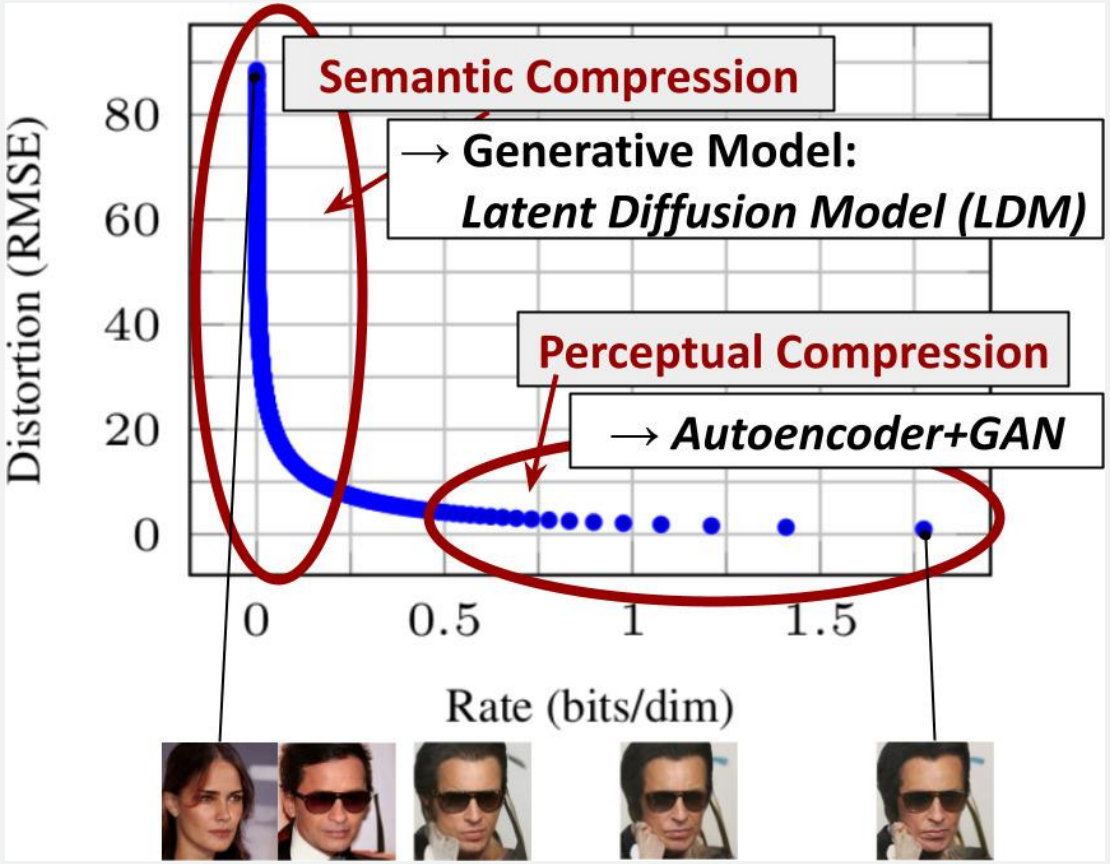


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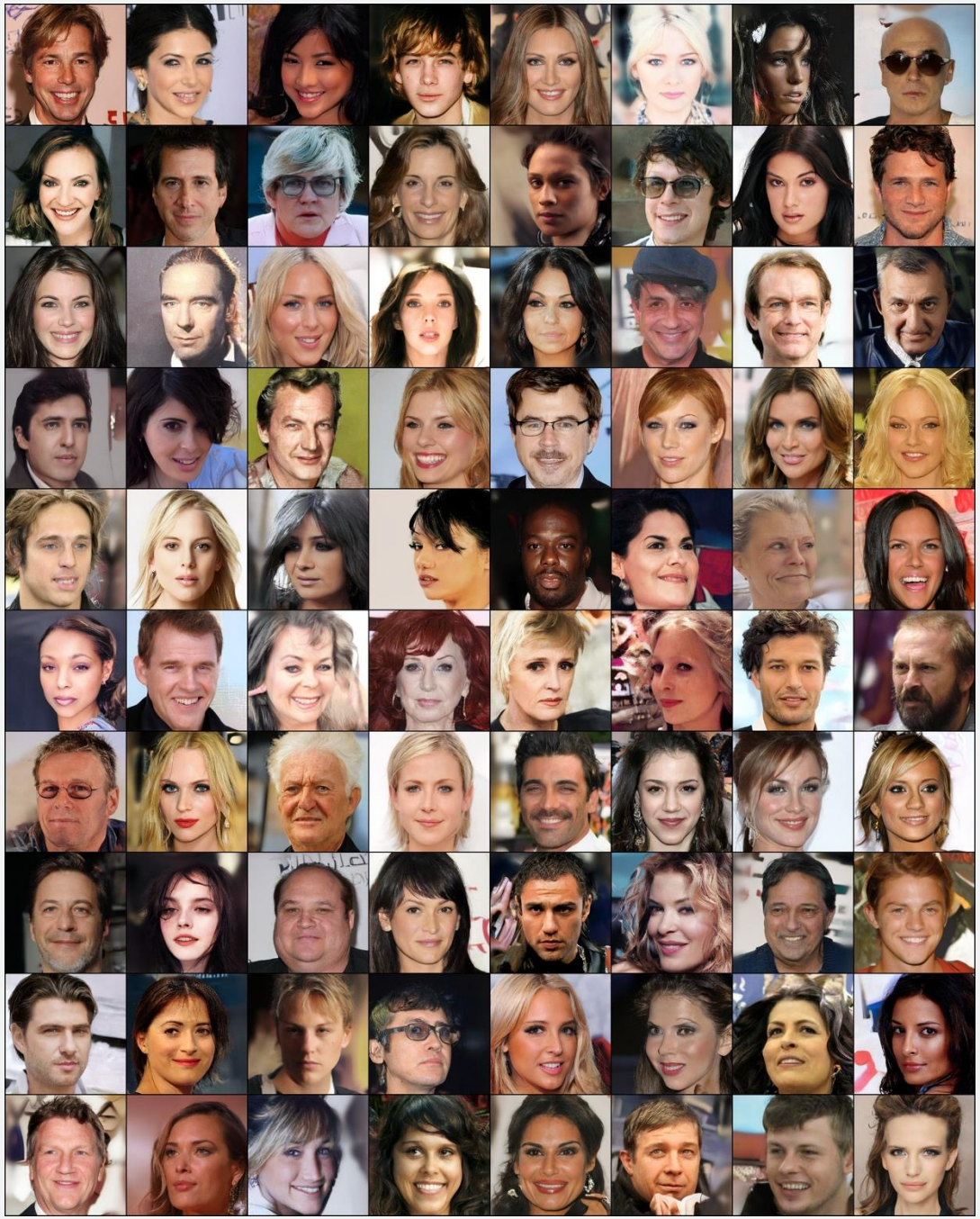
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Various Diffusion Space



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Diffusion Transformer (DiT)

W. Peebles et al., “Scalable Diffusion Models with Transformers”, ICCV 2023 Oral.

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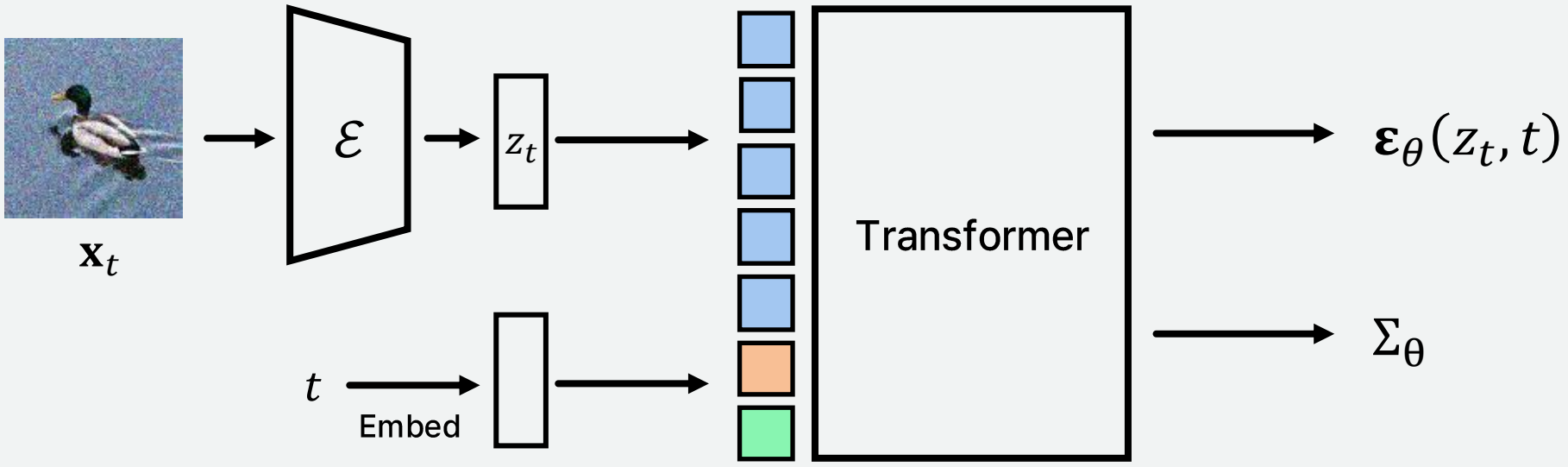
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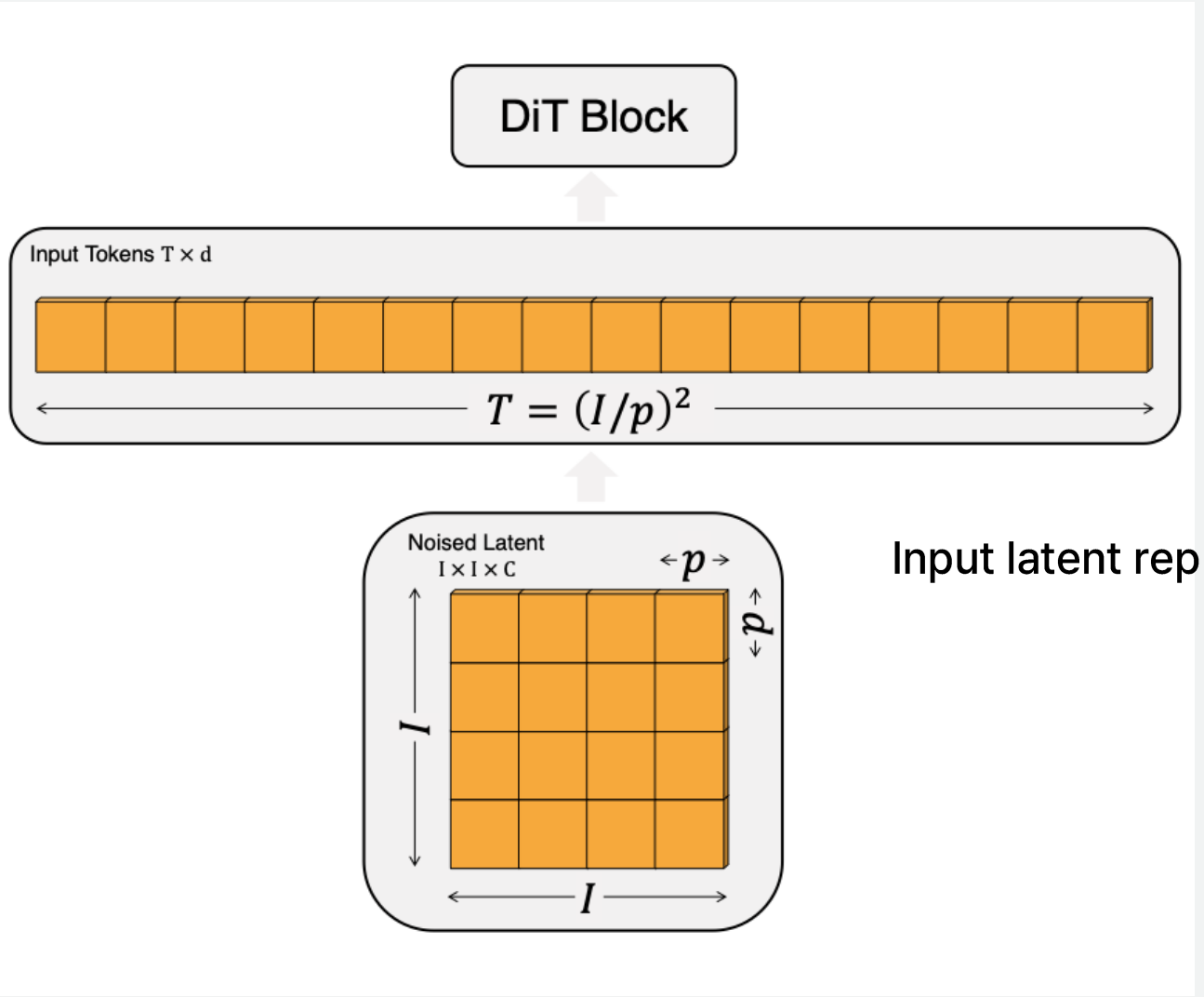
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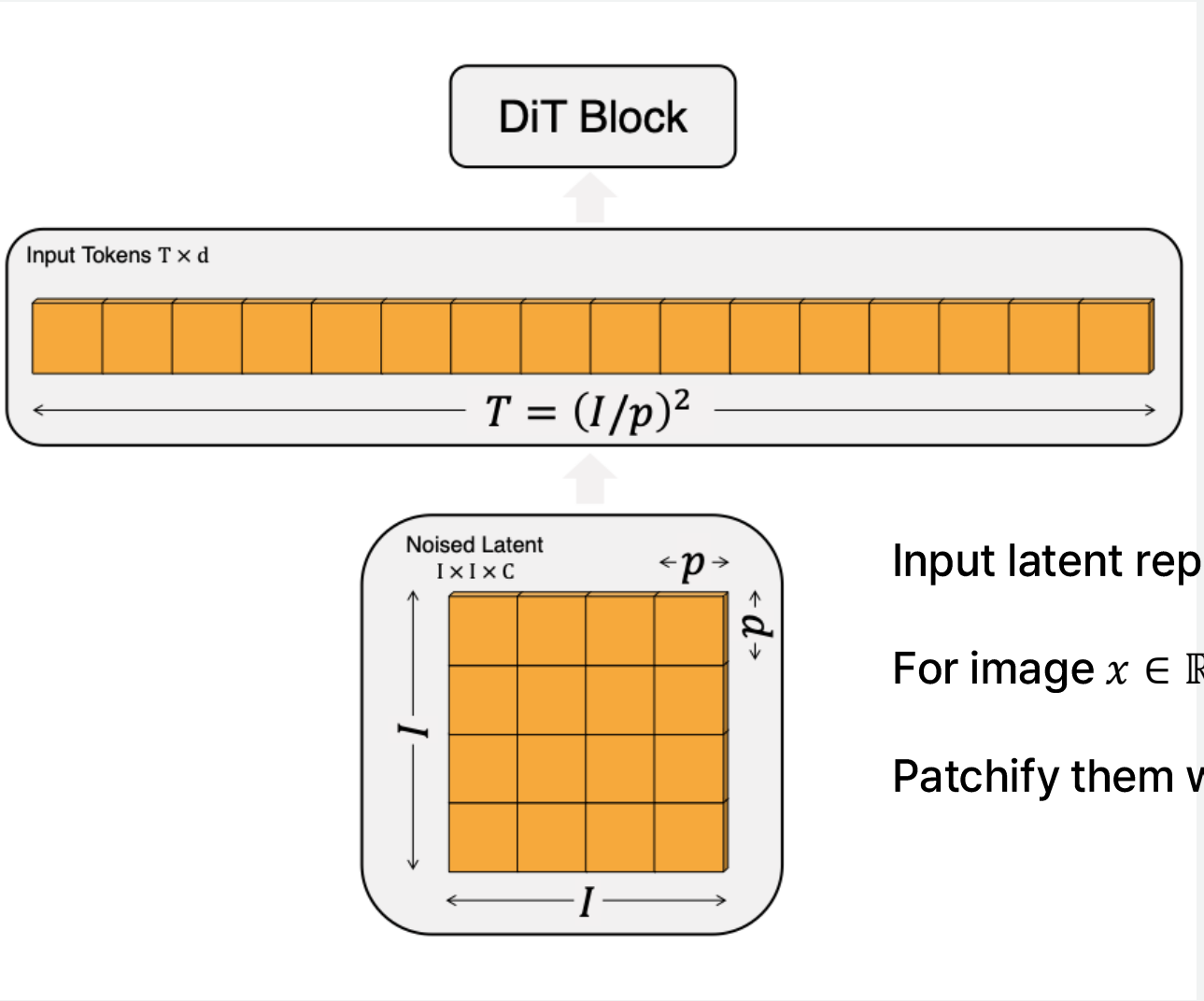
DiT Patchification



Input latent representation $z \in \mathbb{R}^{\mathrm{I}\times\mathrm{I}\times\mathrm{C}}$

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DiT Patchification



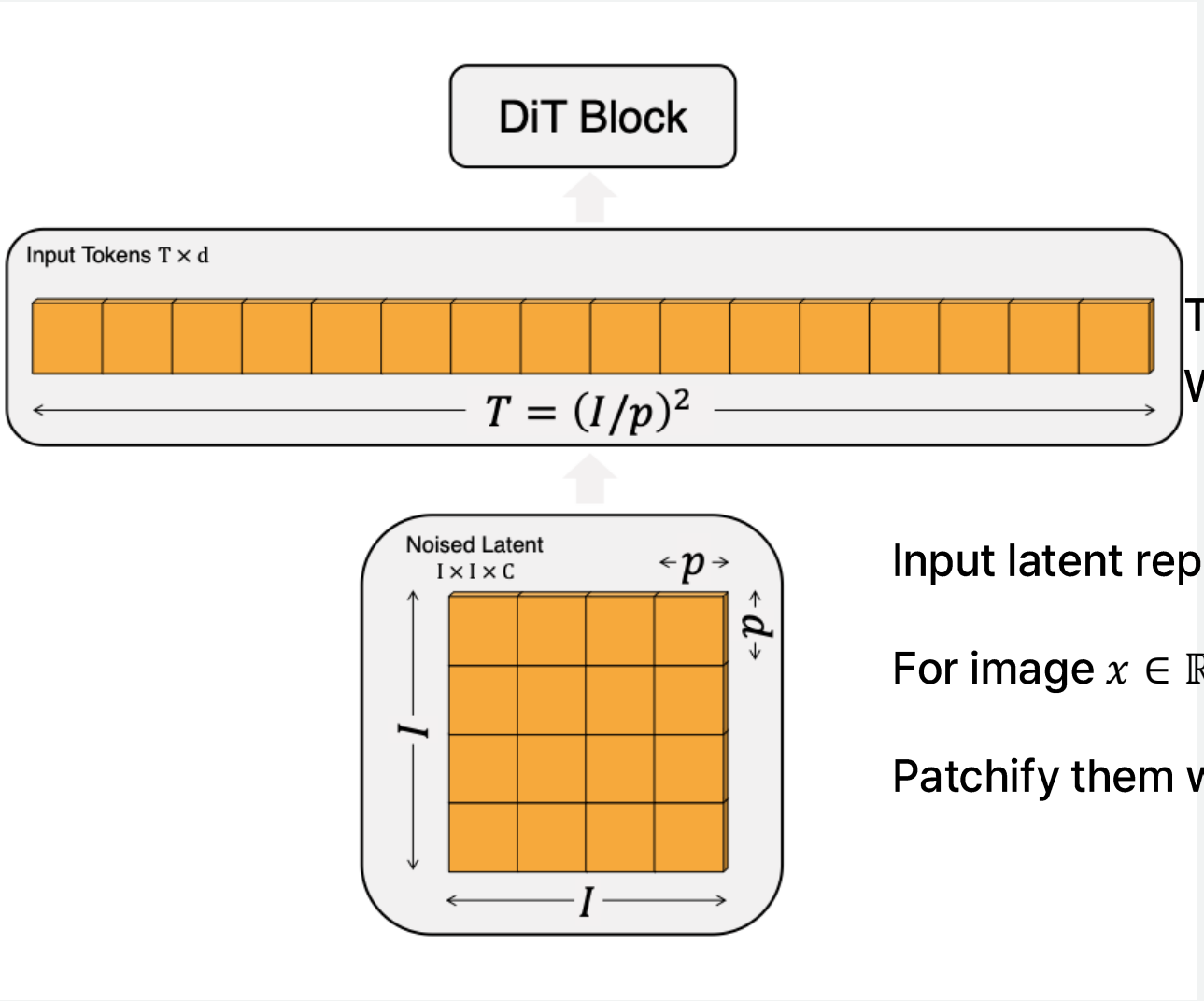
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For image $x \in \mathbb{R}^{256\times256\times3}$, $z \in \mathbb{R}^{32\times32\times4}$

Patchify them with patch size p

W. Peebles et al., “Scalable Diffusion Models with Transformers”, ICCV 2023 Oral.

DiT Patchification



The result is $T = \left(\frac{I}{p}\right)^2$ tokens

With each size d

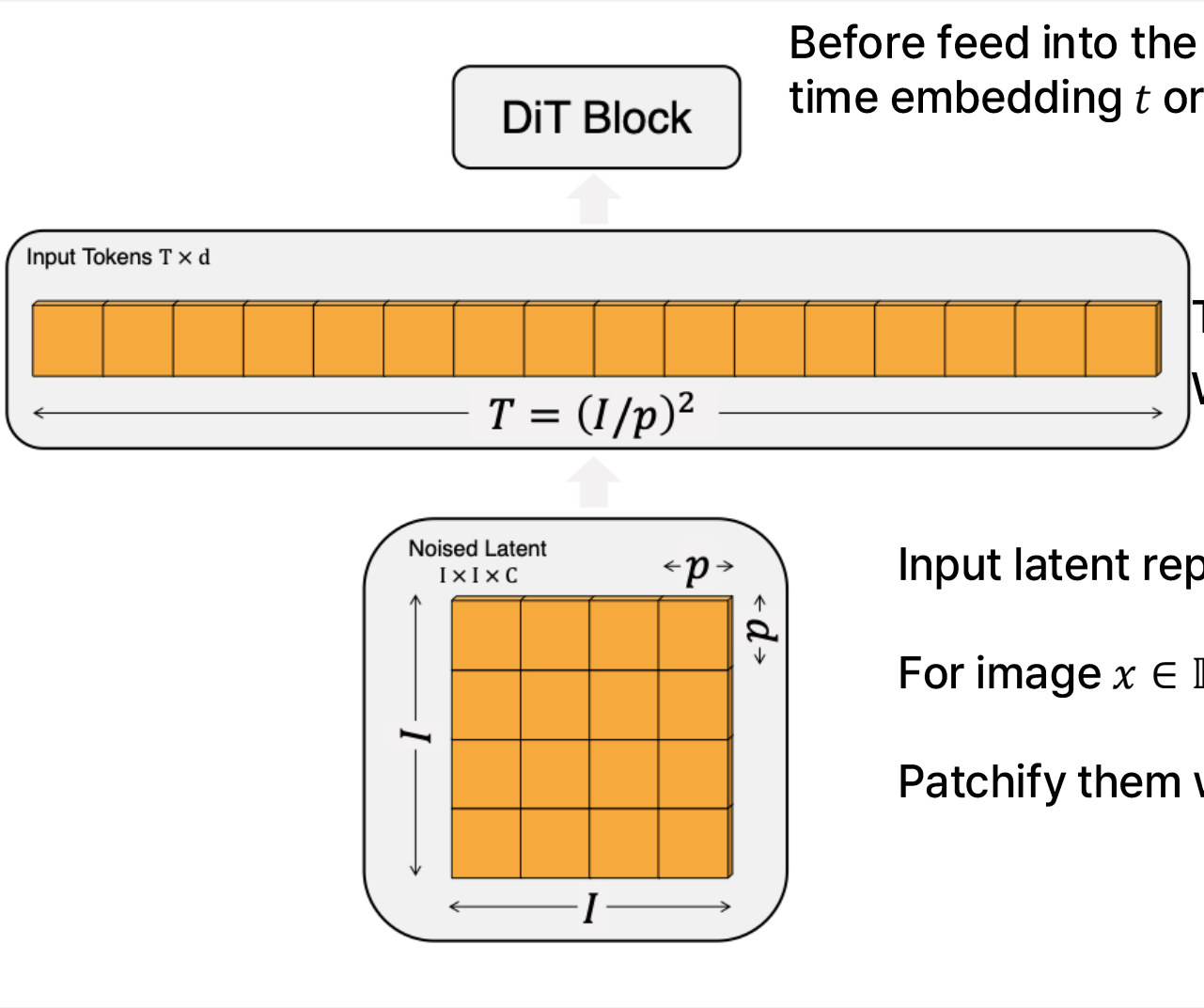
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DiT Patchification



Before feed into the DiT block,
time embedding t or condition c are appended

The result is $T = \left(\frac{I}{p}\right)^2$ tokens

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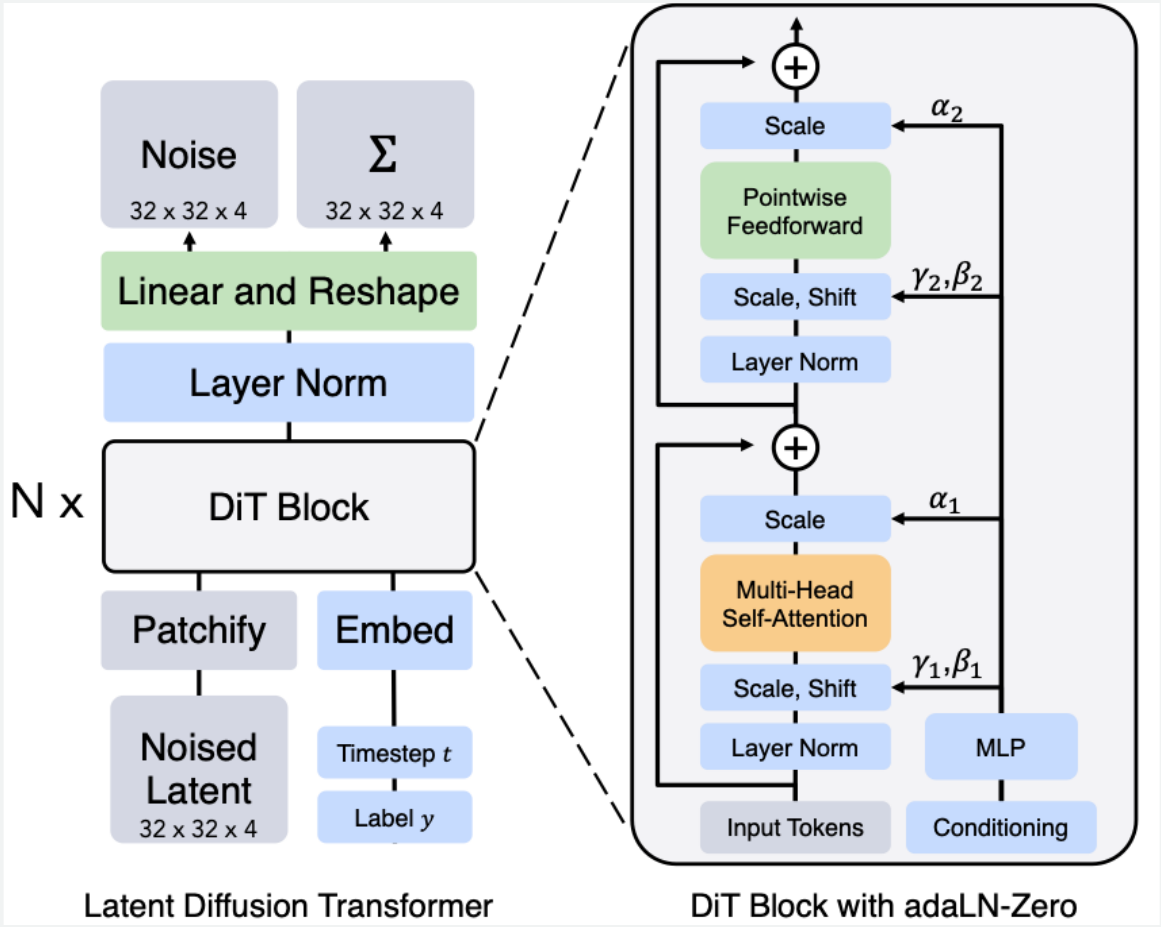
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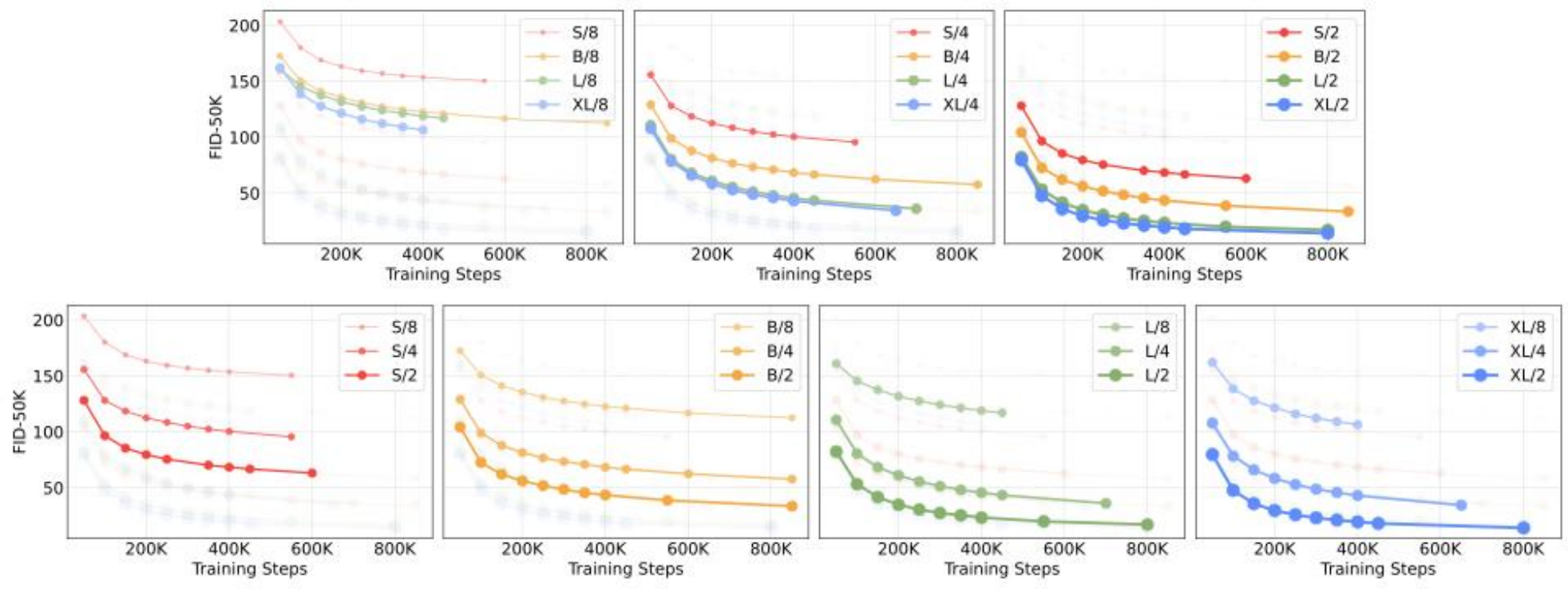
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Architectural Detail



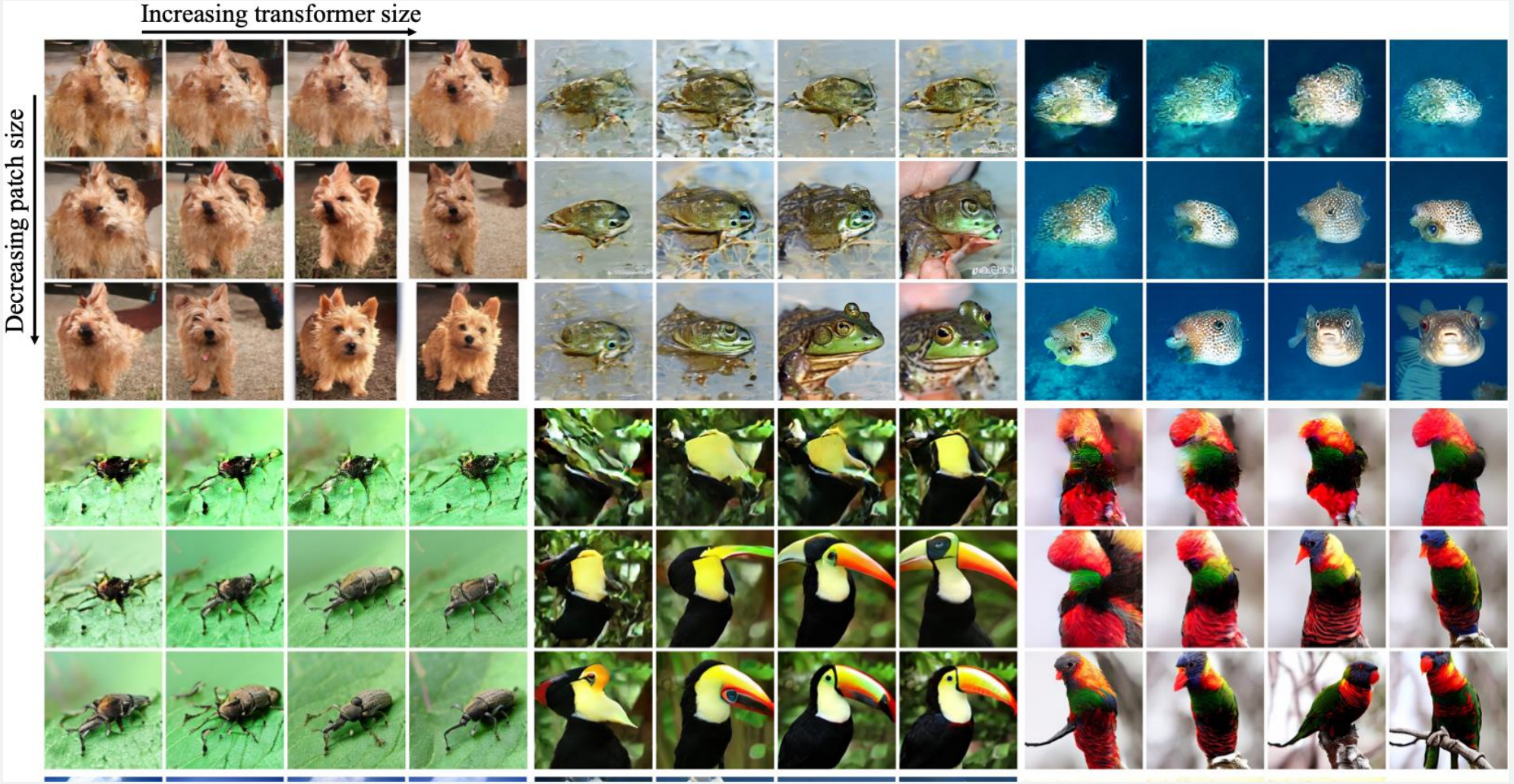
W. Peebles et al., "Scalable Diffusion Models with Transformers", ICCV 2023 Oral.

DiT Scalability



W. Peebles et al., “Scalable Diffusion Models with Transformers”, ICCV 2023 Oral.

DiT



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Score-based Diffusion Process

- Recall (week 5): diffusion process as SDE

$$dx(t) = -\frac{1}{2}\beta(t)x(t)dt + \sqrt{\beta(t)}d\omega(t)$$

A. Vahdat et al., “Score-based Generative Modeling in Latent Space”, NeurIPS 2021.

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- Reverse-time SDE to backward process

$$dx(t) = [-\frac{1}{2}\beta(t)x(t) - \beta(t)\nabla_x \log q_t\big(x(t)\big)]dt + \sqrt{\beta(t)}d\bar{\omega}(t)$$

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$$\min_{\theta} \mathbb{E}_{t\sim\mathcal{U}[0,T]}\left[\mathbb{E}_{x_0\sim q(x_0)}\mathbb{E}_{x_t\sim q(x_t|x_0)}\left[\left\|\nabla_{x_t}\log q_t(x_t|x_0) - s_\theta(x_t,t)\right\|_2^2\right]\right]$$

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However, using this objective is not possible;
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However, using this objective is not possible;
marginal score $\nabla_{z_t}\log q_t(z_t|z_0)$ is non-normal distribution
-> Not able to analytically compute the value

A. Vahdat et al., “Score-based Generative Modeling in Latent Space”, NeurIPS 2021.

LSGM Loss

- Goal: Use the score function $\nabla_{z_t} \log p_\theta(z_t)$ to sample z_0

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Recall (week 2): Variational Lower Bound

$$= \mathbb{E}_{z\sim q_\phi}\left[\log p_\psi(x|z)\right] - D_{KL}\left(q_\phi(z|x)||p(z)\right) + D_{KL}\left(q_\phi(z|x)||p_\psi(z|x)\right)$$

Reconstruction term Prior / Encoder KL Divergence Intractable

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Reconstruction term — Prior / Encoder KL Divergence — Intractable

We have three components:

A. Vahdat et al., "Score-based Generative Modeling in Latent Space", NeurIPS 2021.

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Reconstruction term | Prior / Encoder KL Divergence | Intractable

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Reconstruction term Prior / Encoder KL Divergence Intractable

We have three components:

- Encoder $q_\phi(z_0|x)$
- Decoder $p_\psi(x|z_0)$

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Recall (week 2): Variational Lower Bound

$$= \mathbb{E}_{z\sim q_\phi}\left[\log p_\psi(x|z)\right] - D_{KL}\left(q_\phi(z|x)||p(z)\right) + D_{KL}\left(q_\phi(z|x)||p_\psi(z|x)\right)$$

Reconstruction term | Prior / Encoder KL Divergence | Intractable

We have three components:

- Encoder $q_\phi(z_0|x)$
- Decoder $p_\psi(x|z_0)$
- SGM prior $p_\theta(z_0)$

A. Vahdat et al., "Score-based Generative Modeling in Latent Space", NeurIPS 2021.

LSGM Loss

- Encoder $q_\phi(z_0|x)$
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- Goal: Use the score function $\nabla_{z_t} \log p_\theta(z_t)$ to sample z_0

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$$\mathcal{L}(x, \phi, \theta, \psi)$$

$$= \mathbb{E}_{q_\phi(z_0|x)}\left[-\log p_\psi(x|z_0)\right] + D_{KL}\left(q_\phi(z_0|x)||p_\theta(z_0)\right)$$

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$$D_{KL}(q, p) = \int q \log \frac{q}{p}$$
$$= \int q \log q - \int q \log p$$
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Reconstruction term | Negative encoder entropy | Cross entropy

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Estimated by encoder and decoder

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How do we optimize this?

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LSGM Loss: The Cross Entropy Term

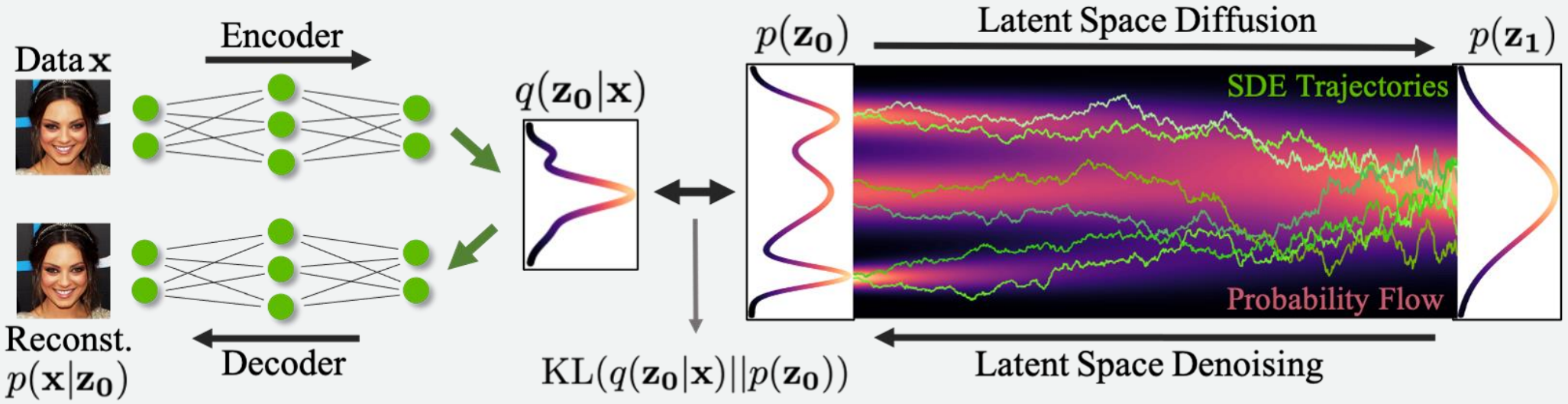
- The cross-entropy term is equivalent to:

$$CE\big(q(z_0|x)||p(z_0)\big)$$

$$= \min_{\theta} \mathbb{E}_{t\sim\mathcal{U}[0,T]}\left[\frac{g(t)^2}{2}\mathbb{E}_{z_t\sim q(z_t,z_0|x)}\left[\left\|\nabla_{z_t}\log q_t(z_t|z_0) - s_\theta(z_t,t)\right\|_2^2\right]\right] + \frac{D}{2}\log(2\pi e\sigma_0^2)$$

A. Vahdat et al., “Score-based Generative Modeling in Latent Space”, NeurIPS 2021.

Score-based Generation in Latent Space



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LSGM Results



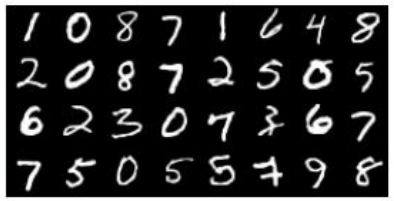
(a) CIFAR-10



(b) CelebA-HQ-256



(c) OMNIGLOT



(d) MNIST

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Discrete Diffusion Models

Discrete Space Transition

- When the output space is discrete, forward pass can be interpreted as transition

J. Austin et al., “Structured Denoising Diffusion Models in Discrete State-Spaces”, NeurIPS 2021.

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$$[Q_t]_{ij} = q(x_t = j | x_{t-1} = i)$$

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Discrete Space Transition

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$$[Q_t]_{ij} = q(x_t = j | x_{t-1} = i)$$

When q is a one-hot vector, then it can be written as follows:

$$q(x_t | x_{t-1}) = \text{Categorical}(x_t; p = x_{t-1} Q_t)$$

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Discrete Space Transition

- T step marginal and posterior at time t-1

$$q(x_t|x_0) = \text{Categorical}(x_t; p = x_0\overline{Q_t}) \quad \text{with} \quad \overline{Q_t} = Q_1 Q_2 \cdots Q_t$$

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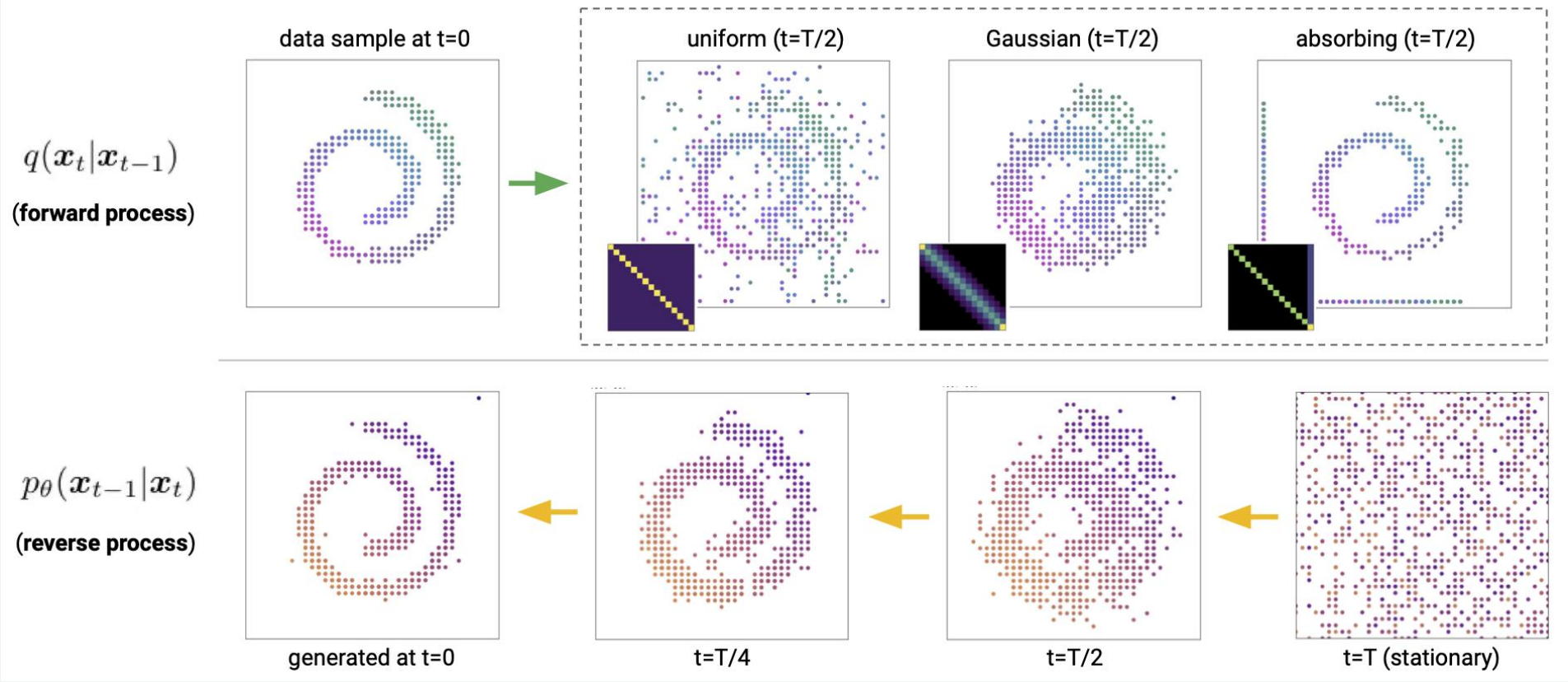
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$$q(x_{t-1}|x_t, x_0) = \frac{q(x_t|x_{t-1}, x_0)q(x_{t-1}|x_0)}{q(x_t|x_0)}$$

$$= \text{Categorical}\left(x_{t-1}; p = \frac{x_t Q_t^\top \odot x_0 \overline{Q_{t-1}}}{x_0 \overline{Q_t} x_t^\top}\right)$$

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Transition Matrices



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Everything is a Diffusion Model

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Everything is a Diffusion Model

- **BERT is a one-step diffusion model**
  - If D3PM uses a uniform transition matrix and an absorbing state at the [MASK] token
  - BERT denoising objective: $\mathcal{L}_{BERT} = \mathcal{L}_{vb} - \mathcal{L}_T = -\mathbb{E}_{q(x_1|x_0)}[\log p_\theta(x_0|x_1)$

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- **Autoregressive models are diffusion models**
  - If masks are deterministically selected with length $N = T$: $q([x_t]_i|x_0) = [x_0]_i$ if $i < N - t$ else [MASK]

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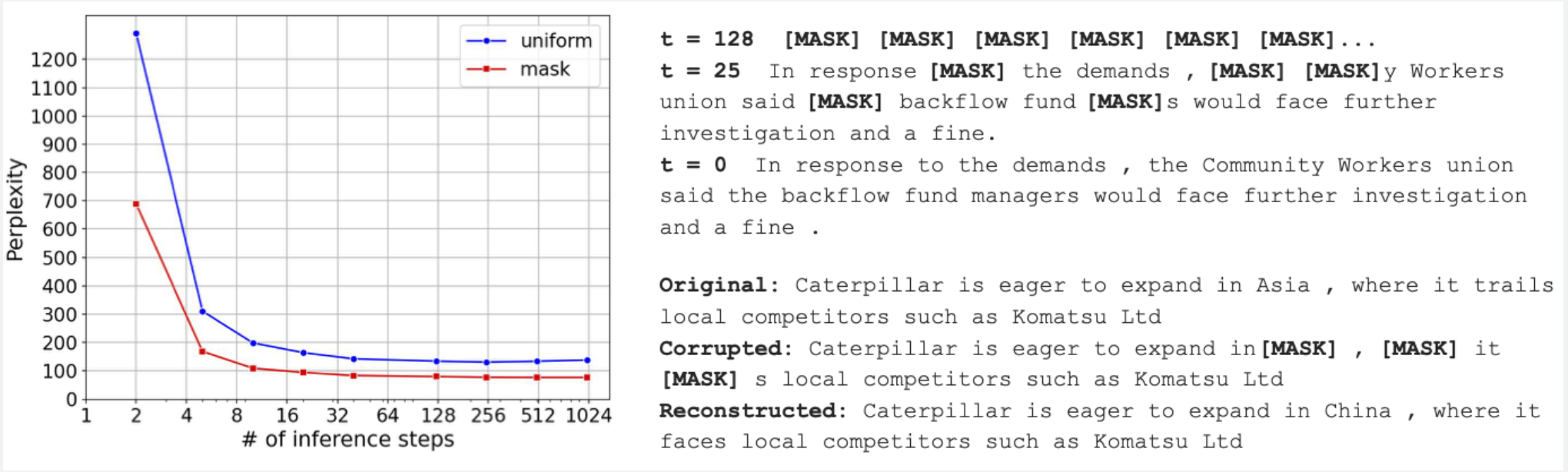
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- **Masked Language-Models are diffusion models**
  - D3PM with an absorbing state is equivalent to the MLM objective

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D3PM Language Generation

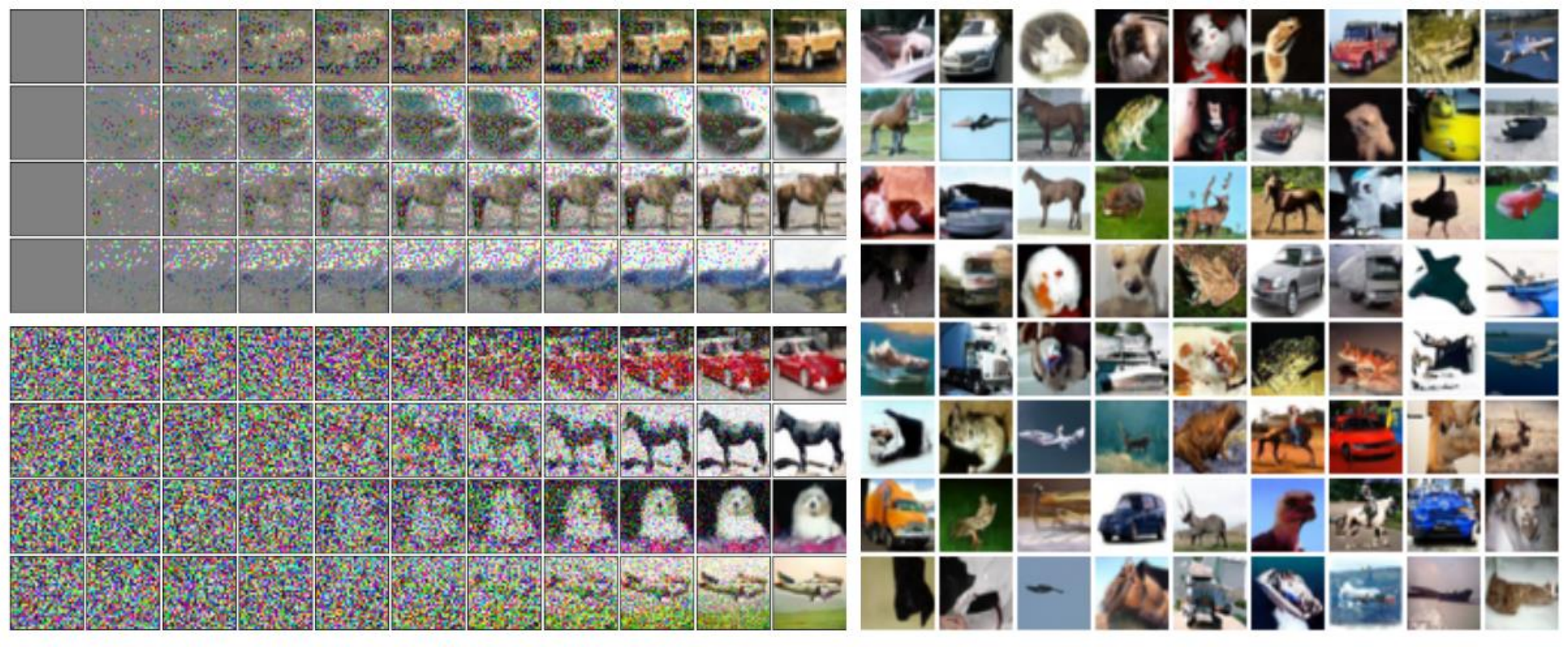


t = 128 **[MASK] [MASK] [MASK] [MASK] [MASK] [MASK]...**
t = 25 In response **[MASK]** the demands , **[MASK] [MASK]**y Workers union said **[MASK]** backflow fund **[MASK]**s would face further investigation and a fine.
t = 0 In response to the demands , the Community Workers union said the backflow fund managers would face further investigation and a fine .

Original: Caterpillar is eager to expand in Asia , where it trails local competitors such as Komatsu Ltd
Corrupted: Caterpillar is eager to expand in**[MASK]** , **[MASK]** it **[MASK]** s local competitors such as Komatsu Ltd
Reconstructed: Caterpillar is eager to expand in China , where it faces local competitors such as Komatsu Ltd

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D3PM Image Generation



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VQ-Diffusion

- Goal: Generate image using discrete diffusion model

S. Gu et al., “Vector Quantized Diffusion Model for Text-to-Image Synthesis”, CVPR 2022 Oral.

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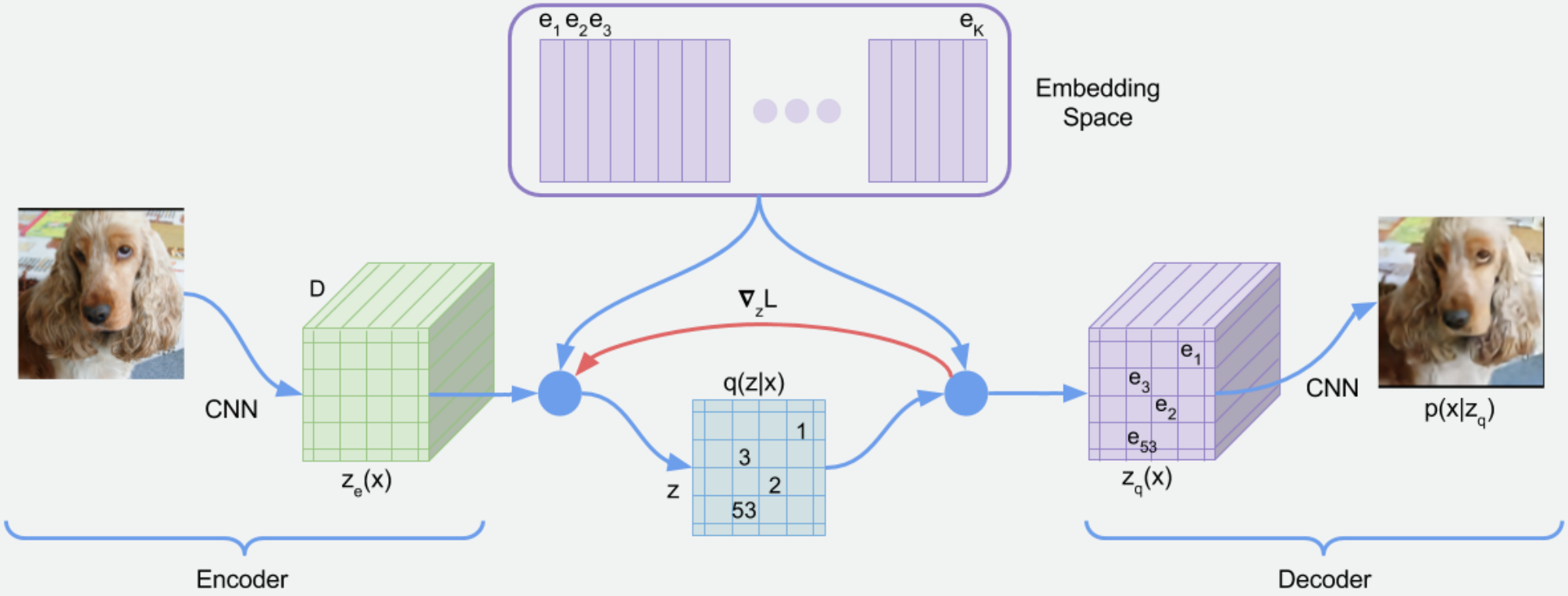
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  - Use discrete representation of image latent space

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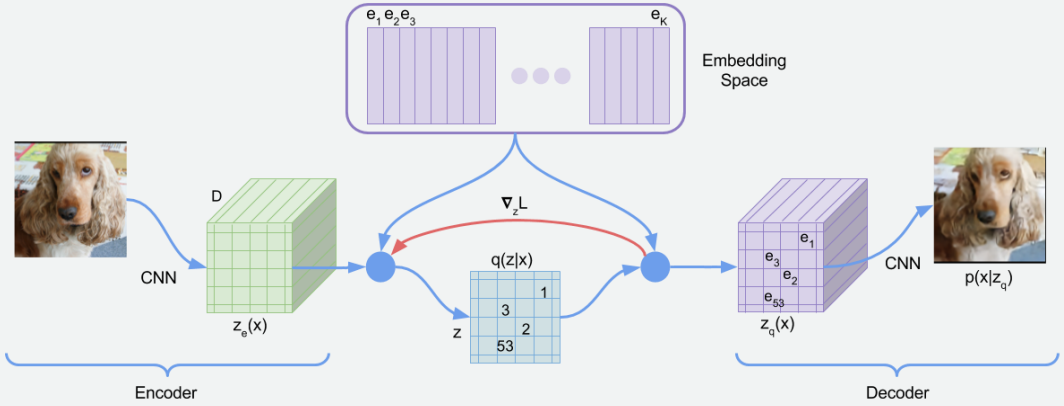
Prerequisites: VQ-VAE



A. van den Oord et al., "Neural Discrete Representation Learning", NIPS 2017.

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- An image $x \in \mathbb{R}^{H \times W \times 3}$ is encoded into a code e_k

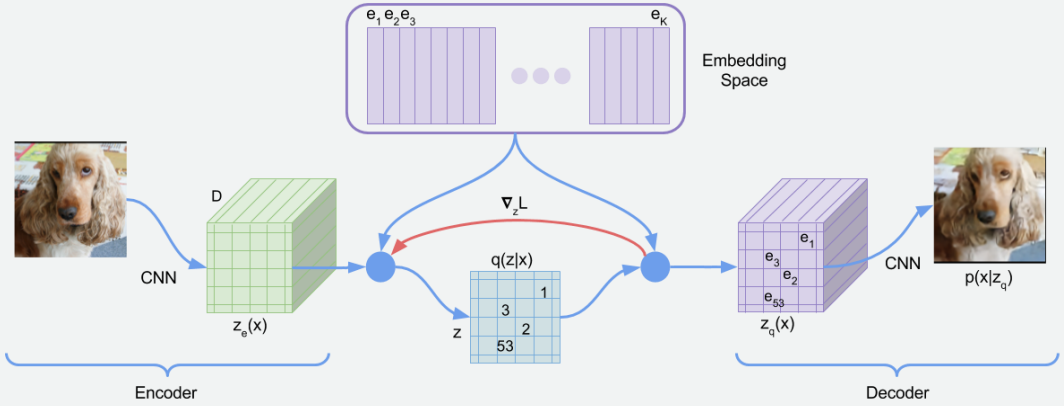


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$$q(z = k|x) = \begin{cases} 1 & \text{for } k = \arg\min_j \left\| z_e(x) - e_j \right\|_2^2 \\ 0 & \text{otherwise} \end{cases}$$



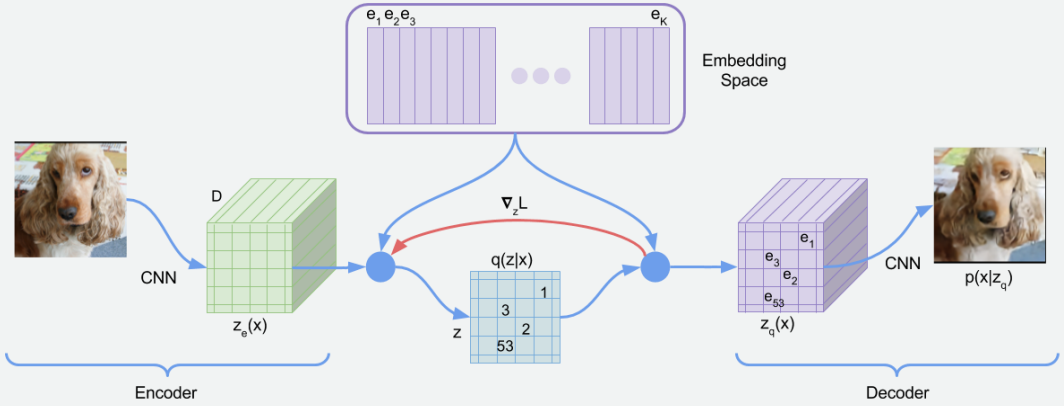
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Vector Quantization

- Goal: Generate image
  - Training process discrepancy: x_0 is not given in the generation process

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Vector Quantization

- Goal: Generate image from text
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  - Use text y as a condition and optimize $p(x|y)$

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Vector Quantization

- Goal: Generate image from text
  - Training process discrepancy: x_0 is not given in the generation process
  - Use text y as a condition and optimize $p(x|y)$
- The objective function can be defined as:

$$\mathcal{L} = \log p_\theta(x_0|x_1, y)$$

$$-\sum_{t=1}^{T-1} D_{KL}\left(q_\phi(x_{t-1}|x_t, x_0)||p(x_{t-1}|x_t, y)\right)$$

$$+D_{KL}\left(q_\phi(x_T|x_0)||p_\psi(x_T)\right)$$

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Mask-and-Replace

- D3PM uses random changing masking strategy:

$$\boldsymbol{Q}_t = \begin{bmatrix} \alpha_t + \beta_t & \beta_t & \cdots & \beta_t \\ \beta_t & \alpha_t + \beta_t & \cdots & \beta_t \\ \vdots & \vdots & \ddots & \vdots \\ \beta_t & \beta_t & \cdots & \alpha_t + \beta_t \end{bmatrix}$$

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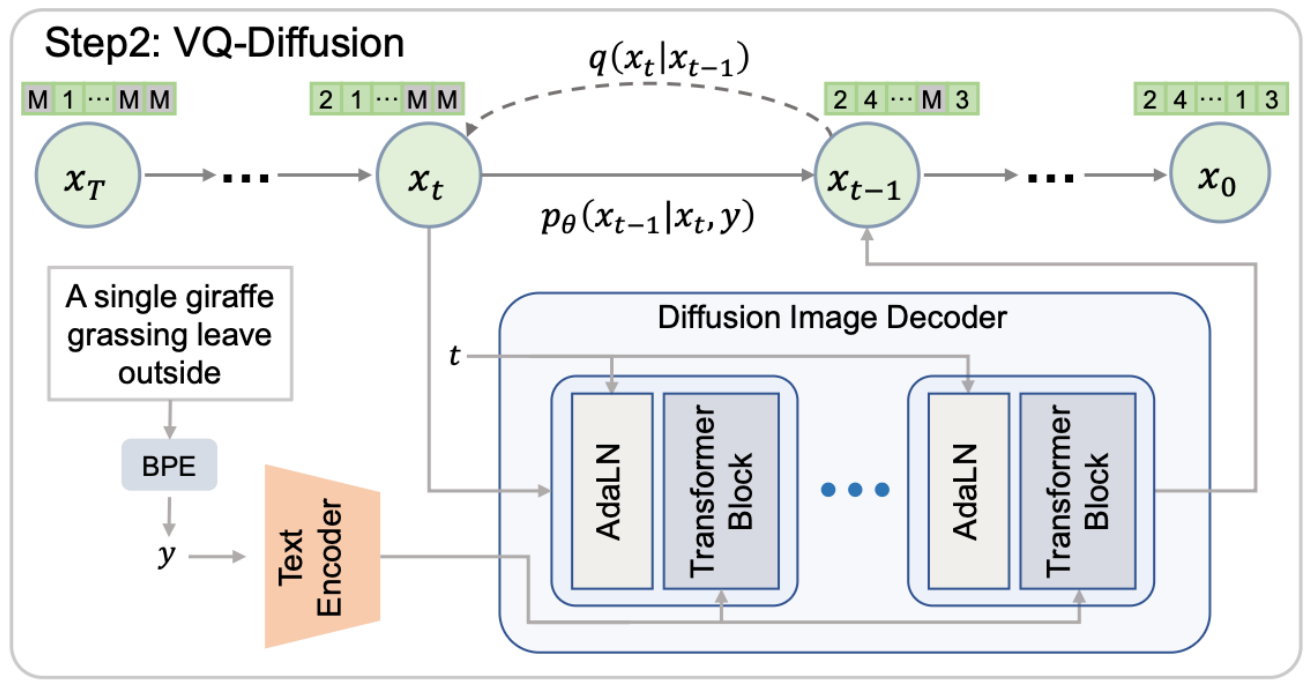
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- Problem: the model cannot know which tokens are masked
- Solution: add a new [MASK] token

$$\boldsymbol{Q}_t = \begin{bmatrix} \alpha_t + \beta_t & \beta_t & \beta_t & \cdots & 0 \\ \beta_t & \alpha_t + \beta_t & \beta_t & \cdots & 0 \\ \beta_t & \beta_t & \alpha_t + \beta_t & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \gamma_t & \gamma_t & \gamma_t & \cdots & 1 \end{bmatrix}$$

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VQ-Diffusion Architecture



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VQ-Diffusion Results



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Thank you!

Jihwan Hong
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